

Mars Slope and Landing Site Analysis

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Abstract

This project presents a GIS based site suitability analysis for Mars using ArcGIS Pro, leveraging elevation data from the MOLA DEM data available by NASA. Slope was extracted from the DEM and used as the primary suitability criterion, identifying regions of low slope that are favorable for safe landings and operations. All major historical Mars landing sites were georeferenced and mapped as reference points, giving spatial context for site evaluation and patterns across missions.

Methods

MOLA DEM data was acquired and imported as the elevation dataset. A slope raster was derived from the DEM and reclassified into suitability categories, with lower slope values assigned a higher suitability scores. Mars landing sites were digitized as points with the name labeled. All layers were assigned to Mars 2000 (Sphere) coordinate system.

Findings

The slope-based suitability analysis showed that large portions of the Martian northern lowlands and equatorial plains rank as the most favorable terrain for landing, with a consistent flag and low elevation character of these regions. Mapping previous landing sites alongside the surface confirms this agreement. The spatial pattern points to mission site selections being target towards flatter terrain



RUTGERS



Legend

